

OBJECT-ORIENTED SOFTWARE ENGINEERING

Assignment 2 – Complete Solution

Project: Smart Waste Management System

Prepared using the Smart Waste Management System diagrams from Assignment 1 and the formulas/methods provided in Unit 2.

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Subject: Object-Oriented Software Engineering

Source basis used for this report

- Assignment 2 question paper: OOSE_Assignment 2.pdf
- Assignment 1 solution diagrams: Assignment1_14_24.pdf (Use Case, Activity, Sequence and Class diagrams)
- Unit 2 course material: 2.1_OOSE_Unit 2.pdf and 2.2_OOSE_Unit 2.pdf

1. Introduction

The Smart Waste Management System is a municipal platform for bin monitoring, waste reporting, collection scheduling, vehicle tracking, worker allocation, complaint management, notifications and operational dashboards. The Assignment 1 solution also includes Computer Vision/ML for waste-category identification and collection-demand prediction.

The estimation below uses the same system modeled in Assignment 1. The use-case diagram supplies actors and use cases; the activity and sequence diagrams clarify workflow and external interactions; and the class diagram supplies the classes, attributes and public methods required for the class-based estimates.

2. Assumptions Used for Estimation

Assumption	Value / Rationale
Working time	8 hours per person-day and 20 person-days (160 hours) per person-month.
Lorenz–Kidd interface	Complex GUI. The system contains dashboards, maps/GPS tracking, image upload, multi-role workflows and interactive operational views.
Lorenz–Kidd productivity	18 person-days per class, within the Unit 2 suggested 15–20 PD/class range; selected because the system includes AI/ML and multiple service integrations.
UCP productivity	20 person-hours/UCP, the historical/default value presented in Unit 2.
Class Point productivity	6 person-hours per adjusted class point, matching the worked example in Unit 2.
OOFp productivity	8 person-hours/function point, matching the worked example in Unit 2.
Adjustment factors	Ratings are project-specific analytical assumptions derived from the Smart Waste Management System requirements and diagrams. They are shown explicitly rather than hidden.

3. Project Estimation

3.1 Lorenz and Kidd Estimation Method

Unit 2 defines Lorenz–Kidd as a class-counting method that separates **key classes** derived directly from the problem domain from implementation/support classes. Support classes are estimated by multiplying key classes by an interface complexity multiplier, and effort is then obtained from total classes × person-days per class.

Formula 1: Support Classes = Key Classes × Interface Multiplier

Formula 2: Total Classes = Key Classes + Support Classes

Formula 3: Total Effort = Total Classes × Person-Days per Class

Key classes identified from Assignment 1 class diagram: 18

Key/domain class	Reason
User hierarchy: User, Citizen, MunicipalOfficer, Driver, Worker, Supervisor, Administrator	Represents system users and roles explicitly modeled in Assignment 1.
WasteBin	Municipal waste bin being monitored.
WasteReport	Citizen-generated waste/full-bin report.
WasteClassification	Result of computer-vision waste classification.
CollectionSchedule	Generated and approved collection plan.
CollectionRequest	Collection demand/request with priority and status.
Vehicle	Collection vehicle managed and assigned by the system.
GPSLocation	Vehicle/location tracking data.
WorkerAssignment	Assignment of waste workers.
Complaint	Citizen complaint and resolution workflow.

Key/domain class	Reason
WasteCollection	Actual collection record.
DemandPrediction	ML-generated collection-demand prediction.

The remaining implementation-oriented classes in the diagram—NotificationService, TrackingService, MLService and AuditLog—support the architecture. Under the Lorenz–Kidd procedure, support-class effort is estimated using the interface multiplier rather than merely counting those four diagram classes.

Calculation	Result
Key classes	18
Interface type and multiplier	Complex GUI → 3.0
Support classes	$18 \times 3.0 = 54$
Estimated total classes	$18 + 54 = 72$
Productivity	18 person-days/class
Effort	$72 \times 18 = 1,296$ person-days
Person-hours	$1,296 \times 8 = 10,368$ hours
Person-months	$1,296 \div 20 = 64.80$ person-months

3.2 Use Case Point (UCP) Method

Unit 2 gives the UCP formula as $UCP = (UAW + UUCW) \times TCF \times ECF$. Actors are weighted as Simple = 1, Average = 2 and Complex = 3. Use cases are weighted as Simple = 5, Average = 10 and Complex = 15.

Formula: $UCP = (UAW + UUCW) \times TCF \times ECF$

Effort: Person-hours = $UCP \times \text{Productivity Factor}$

3.2.1 Unadjusted Actor Weight (UAW)

Actor	Classification	Weight
Citizen	Complex human GUI actor	3
Municipal Officer	Complex human GUI actor	3
Driver	Complex human GUI actor	3
Waste Worker	Complex human GUI actor	3
Supervisor	Complex human GUI actor	3
Administrator	Complex human GUI actor	3
Computer Vision / ML	Simple external system/API actor	1
Total UAW		$6 \times 3 + 1 \times 1 = 19$

3.2.2 Unadjusted Use Case Weight (UUCW)

The Assignment 1 use-case diagram contains the following functional groups. Transaction complexity is estimated from the interaction implied by the use-case, activity and sequence diagrams. This is an analytical assumption because the original diagram does not provide full textual transaction specifications.

Complexity	Representative use cases	Count	Weight	Subtotal
Simple (≤ 3 transactions)	Register/Login; Track Collection; View Notifications; View Assigned Route; View Work Assignment; View Analytics; Dashboard; Audit Logs; Update Vehicle Status; Notifications	10	5	50
Average (4–7)	Upload Waste Image; Submit Complaint; Manage Bins; Assign Vehicles; Allocate Workers; Monitor Operations; Manage Complaints; Start/Complete Collection; Collect Waste; Update Collection Status; Approve Schedule; Share GPS Location; Monitor Workers/Vehicles; Resolve Escalated Complaints	14	10	140
Complex (> 7)	Report Full Bin/Waste; Manage Collection Schedule; Manage Users/Roles; System Configuration; Waste Category Identification; Collection Demand Prediction; Vehicle Tracking; Centralized System Controls	8	15	120
Total UUCW				310

3.2.3 Technical Complexity Factor (TCF)

The Unit 2 TCF formula is $TCF = 0.6 + 0.01 \times \Sigma(T_i)$. The following ratings (0–5) reflect the distributed, AI-assisted, GPS-enabled and security-sensitive nature of the project.

Technical factor	Weight	Rating	Weighted value
Distributed system	2.0	5	10.0
Response time / performance	1.0	4	4.0
End-user efficiency	1.0	4	4.0
Complex internal processing	1.0	5	5.0
Reusable code target	1.0	2	2.0
Easy to install	0.5	3	1.5
Easy to use	0.5	4	2.0
Portable	2.0	3	6.0
Easy to change	1.0	4	4.0
Concurrency	1.0	4	4.0
Security	1.0	5	5.0
Third-party/service integration	1.0	3	3.0
Training needs	1.0	2	2.0
Σ Technical factors			52.5

$$TCF = 0.6 + 0.01 \times 52.5 = 1.125$$

3.2.4 Environmental Complexity Factor (ECF)

Unit 2 uses $EF = 1.4 - 0.03 \times \Sigma(E_i)$, with eight team/project maturity factors. Ratings below are explicit project assumptions.

Environmental factor	Weight	Rating	Weighted value
Familiarity with development process	1.5	3	4.5
Application experience	0.5	3	1.5
OO experience of team	1.0	4	4.0
Lead analyst capability	0.5	4	2.0
Team motivation	1.0	4	4.0
Stable requirements	2.0	3	6.0
Part-time staff	-1.0	2	-2.0
Difficult programming language	-1.0	2	-2.0
Σ Environmental factors			18.0

$$ECF = 1.4 - 0.03 \times 18 = 0.86$$

3.2.5 UCP Result and Effort

Calculation	Result
UAW	19

Calculation	Result
UUCW	310
TCF	1.125
ECF	0.86
UCP	$(19 + 310) \times 1.125 \times 0.86 = 318.31$
Productivity	20 person-hours/UCP
Effort	$318.31 \times 20 = 6,366.15$ person-hours
Person-days	$6,366.15 \div 8 = 795.77$ person-days
Person-months	$795.77 \div 20 = 39.79$ person-months

3.3 Class Point Method

The Unit 2 Class Point method classifies each class as PDT (Problem Domain Type), HIT (Human Interaction Type), DMT (Data Management Type) or TMT (Task Management Type). Complexity is determined using NEM (Number of External Methods) and NSR (Number of Services Requested). The Assignment 1 class diagram gives NEM directly from public methods; NSR is estimated from the class relationships and service dependencies.

Complexity rule used: NEM/NSR ranges 0–3, 4–7 and ≥ 8 determine Low/Average/High according to the Unit 2 CP1 matrix.

Class group / class	Type	NEM / NSR basis	Complexity
User, Citizen, Worker, Supervisor, Administrator, WasteBin, WasteReport, WasteClassification, CollectionRequest, Vehicle, GPSLocation, WorkerAssignment, Complaint, WasteCollection	PDT	Mostly ≤ 3 methods and limited service requests	Low (14)
MunicipalOfficer, Driver, CollectionSchedule, DemandPrediction	PDT	Higher method/service interaction or workflow coupling	Average (4)
NotificationService	TMT	Single exposed operation; limited coupling	Low (1)
TrackingService, MLService	TMT	Multiple operations and external/service interactions	Average (2)
AuditLog	DMT	Persistent/audit data management with no public operation shown	Low (1)

3.3.1 Total Unadjusted Class Points (TUCP)

Type	Low	Average	High	Subtotal
PDT	$14 \times 3 = 42$	$4 \times 6 = 24$	$0 \times 10 = 0$	66
HIT	$0 \times 4 = 0$	$0 \times 7 = 0$	$0 \times 12 = 0$	0
DMT	$1 \times 5 = 5$	$0 \times 8 = 0$	$0 \times 13 = 0$	5
TMT	$1 \times 4 = 4$	$2 \times 6 = 12$	$0 \times 9 = 0$	16
TUCP				87

TUCP = $\Sigma(w_{ij} \times x_{ij}) = 87$ class points

3.3.2 Adjustment and Effort

Unit 2 adjusts the class-point count using $CFP = TUCP \times (0.65 + 0.01 \times \Sigma GSC)$. A project-specific General System Characteristics sum of 48 is assumed because the system is distributed, performance-sensitive, security-sensitive and uses multiple integrations.

Calculation	Result
TUCP	87
ΣGSC	48
VAF	$0.65 + 0.01 \times 48 = 1.13$
Adjusted Class Points (CFP)	$87 \times 1.13 = 98.31$
Productivity	6 person-hours/point
Effort	$98.31 \times 6 = 589.86$ person-hours
Person-days	$589.86 \div 8 = 73.73$ person-days
Person-months	$73.73 \div 20 = 3.69$ person-months

3.4 Object-Oriented Function Point (OOFP) Analysis

Unit 2 maps OO constructs to IFPUG function types: persistent classes → ILF, externally owned referenced classes/files → EIF, create/update/delete methods → EI, derived outputs → EO, and retrieval methods → EQ. The standard weights are ILF/EIF = 7/10/15 and EI = 3/4/6 and EO/EQ = 4/5/7 for Low/Average/High.

3.4.1 Functional Identification

Function type	Project mapping	Complexity/count	Subtotal
ILF	13 persistent logical data groups: User/role data, WasteBin, WasteReport, WasteClassification, CollectionSchedule, CollectionRequest, Vehicle, GPSLocation, WorkerAssignment, Complaint, WasteCollection, DemandPrediction, AuditLog	13 Low × 7	91
EIF	External ML/CV, GPS tracking and notification interfaces	3 Low × 7	21
EI	Create/update operations such as login/logout, report submission, scheduling, assignments, approvals, status/location updates and complaint actions	22 Low × 3 + 6 Average × 4	90
EO	Derived/produced outputs such as dashboards, monitoring output, classification, prediction and notifications	4 Low × 4 + 3 Average × 5	31
EQ	Retrieval functions such as tracking, routes, assignments, audit logs and current location	5 Low × 4 + 1 Average × 5	25
UFP			258

UFP = 91 + 21 + 90 + 31 + 25 = 258

3.4.2 Adjustment and Effort

Calculation	Result
Unadjusted Function Points (UFP)	258
ΣGSC	48
VAF	$0.65 + 0.01 \times 48 = 1.13$
Adjusted Function Points (AFP)	$258 \times 1.13 = 291.54$
Productivity	8 person-hours/FP
Effort	$291.54 \times 8 = 2,332.32$ person-hours
Person-days	$2,332.32 \div 8 = 291.54$ person-days
Person-months	$291.54 \div 20 = 14.58$ person-months

4. Comparison and Analysis

The four methods intentionally produce different values because they measure different aspects of the same project. Unit 2 recommends triangulating estimates rather than trusting a single number.

Method	Primary size result	Person-hours	Person-days	Person-months
Lorenz–Kidd	72 estimated classes	10,368.00	1,296.00	64.80
Use Case Point	318.31 UCP	6,366.15	795.77	39.79
Class Point	98.31 adjusted CP	589.86	73.73	3.69
OOFP	291.54 adjusted FP	2,332.32	291.54	14.58

4.1 Interpretation of Differences

Lorenz–Kidd gives the highest effort because the Complex GUI multiplier expands 18 key classes into 54 estimated support classes and then applies 18 person-days to every estimated class. It is deliberately a coarse early-stage method.

UCP gives the second-highest estimate because the system has many actors and use cases, including complex workflows such as reporting, scheduling, ML prediction and system controls. UCP also reflects technical and environmental difficulty.

Class Point gives the lowest estimate because the existing UML class diagram contains mostly low-to-average complexity classes with relatively few public methods. It measures structural class complexity rather than every end-to-end transaction.

OOF is between UCP and Class Point because individual input, output and inquiry transactions are counted separately, while the persistent data groups are also included.

4.2 Most Appropriate Method

Recommended primary method: Use Case Point (UCP). The Smart Waste Management System already has a detailed use-case model with clearly identifiable human actors and system functionality. The project is strongly use-case driven and includes technical factors such as distribution, performance, security, concurrency and external integrations. Therefore UCP best matches the information available at this stage. OOF should be used as a secondary cross-check because the system has many transactional operations and external interfaces.

5. Risk Analysis

Unit 2 defines risk exposure as $RE = Probability \times Consequence$. Probability is expressed from 0.0 to 1.0, while consequence may be expressed in person-days. Risks are sorted by RE so attention is focused on the highest exposure.

Formula: $RE = P \times C$

Impact values below are expressed as estimated person-days of consequence if the risk occurs. Priority bands used in this report: Critical ≥ 15 , High 10–14.99, Medium 5–9.99, Low < 5 .

ID	Risk	Category	P	Impact (PD)	RE	Priority
R1	Requirements change after estimation	Requirements	0.55	35	19.25	Critical
R2	ML waste classification accuracy is poor	Technical	0.40	45	18.00	Critical
R5	Integration failure among portal, ML, scheduling, GPS and notification services	Technical	0.35	50	17.50	Critical
R6	Security breach or unauthorized role access	Security	0.25	60	15.00	Critical
R3	ML demand prediction is inaccurate	Technical	0.35	40	14.00	High
R11	Key developer or ML specialist unavailable	Resource	0.25	50	12.50	High
R9	Vehicle or worker availability differs from plan	Operational / Resource	0.40	30	12.00	High
R12	Peak workload causes performance problems	Technical	0.30	40	12.00	High
R8	Schedule or approval workflow delays	Project / Operational	0.45	25	11.25	High
R7	Database failure or data loss	Operational / Technical	0.20	55	11.00	High
R4	GPS tracking outage or inaccurate location	Technical	0.30	30	9.00	Medium
R10	Poor network connectivity in the field	Operational	0.45	20	9.00	Medium

5.1 Risk Exposure Matrix

The following matrix uses the same exposure idea (probability $\times$ consequence) and groups the identified risks into practical probability and consequence bands.

Probability \ Impact	Negligible <20 PD	Marginal 20–34 PD	Critical 35–49 PD	Catastrophic ≥ 50 PD
High ≥ 0.50		R1 is near Critical band		
Medium 0.30–0.49	R10 (0.45,20)	R4 (0.30,30); R8 (0.45,25); R9 (0.40,30)	R2 (0.40,45); R3 (0.35,40); R12 (0.30,40)	R5 (0.35,50)
Low < 0.30				R6 (0.25,60); R7 (0.20,55); R11 (0.25,50)

Priority order by risk exposure: R1 (19.25), R2 (18.00), R5 (17.50), R6 (15.00), R3 (14.00), R11 (12.50), R9 (12.00), R12 (12.00), R8 (11.25), R7 (11.00), R4 (9.00), R10 (9.00).

6. RMMM Plan

RMMM means Risk Mitigation, Monitoring and Management. Unit 2 requires each major risk to record its probability/consequence/exposure, mitigation actions, monitoring indicators, contingency plan, owner and trigger/threshold.

Risk	Mitigation	Monitoring	Management / Contingency	Owner / Trigger
R1 Requirements change late	Freeze baseline after SRS approval; prioritize changes; use change control.	Review change-request rate weekly.	Use contingency buffer and Change Control Board; re-estimate affected work.	Project Manager / Trigger: repeated major scope changes.
R2 Poor ML classification accuracy	Validate data, define confidence thresholds, test model before deployment.	Track accuracy, confidence and misclassification rate.	Route low-confidence reports to manual municipal verification.	ML Lead / Trigger: accuracy below agreed threshold.
R5 Service integration failure	Define APIs/contracts early; use integration tests and staging environment.	Track failed API calls, latency and integration test pass rate.	Fallback to manual scheduling/queue/retry; isolate failing service.	Technical Lead / Trigger: repeated integration failures.
R6 Security breach / unauthorized access	RBAC, secure authentication, input validation, audit logging and least privilege.	Monitor failed logins, unusual access and audit events.	Disable affected accounts/sessions, investigate logs, rotate credentials and restore secure state.	Security/Admin Lead / Trigger: suspicious access pattern.
R3 Inaccurate demand prediction	Validate model on historical/holdout data; allow human override.	Monitor prediction error and officer overrides.	Use rule-based or manual priority setting until model is corrected.	ML Lead / Trigger: sustained prediction error.
R11 Key specialist unavailable	Document architecture/model; cross-train team members.	Review staffing availability and knowledge coverage.	Reassign work, use backup specialist and revise schedule.	Project Manager / Trigger: prolonged unavailability.
R9 Vehicle/worker shortage	Maintain availability data and backup resources.	Compare scheduled versus actual availability daily.	Re-optimize routes/schedules and allocate backup resources.	Operations Manager / Trigger: assignment cannot be fulfilled.
R12 Peak performance problems	Load testing, caching, asynchronous queues and capacity planning.	Monitor response time, CPU/database utilization and queue length.	Scale services and temporarily defer non-critical analytics.	Technical Lead / Trigger: SLA/performance threshold exceeded.

7. Conclusion

The Smart Waste Management System was estimated using four object-oriented estimation approaches required by Assignment 2. The estimates differ because each method measures a different project view: classes and interface complexity (Lorenz–Kidd), user interactions and project factors (UCP), structural class complexity (Class Point), and transactional/data functionality (OOF). For this use-case-driven municipal system, UCP is recommended as the primary estimate, with OOF as the strongest functional cross-check.

Risk analysis identified twelve major risks. The highest exposure risks are requirements change, ML classification quality, multi-service integration and security. The RMMM plan converts those risks into concrete mitigation, monitoring and contingency actions.

8. Source References

1. OOSE_Assignment 2.pdf – Assignment requirements and submission tasks.
2. Assignment1_14_24.pdf – Smart Waste Management System Use Case Diagram (page 1), Activity Diagram (page 2), Sequence Diagram (page 3), and Class Diagram (page 4).
3. 2.1_OOSE_Unit 2.pdf – Object-Oriented Software Estimation overview and UCP formula/material.
4. 2.2_OOSE_Unit 2.pdf – Lorenz–Kidd method (slides/pages 3–10), UCP method (12–18), Class Point method (20–25), OOF method (27–31), and Risk Management/RMMM (33–41).

Note: Numerical factor ratings and complexity classifications that are not explicitly supplied by the diagrams are clearly stated analytical assumptions. They are intended to demonstrate application of the Unit 2 formulas to the Assignment 1 project rather than to claim measured historical productivity.